1) Medical Image Enhancement

(i) Negative Transformation.

(ii) It reverses the gray levels, making dark tissues appear brighter while bones become darker. This improves the visibility of darker regions in x-ray images.

(iii) • Dark tissues become brighter and easier to observe.

• Hidden details in soft tissues become more visible.

• Overall image contrast for dark regions is improved.

2) Night-Time Surveillance

(i) Log Transformation.

(ii) A log transformation enhances low intensity (dark) pixel values without increasing the contrast of fingerprint ridges and valleys, making fine ridge details clearer for accurate fingerprint recognition

(iii) → The image becomes brighter.

→ vehicles, people, and objects are more clearly visible.

→ Dark areas gain better contrast while bright areas remain almost unchanged.

3) Fingerprint Recognition.

(i) Power-Law (Gamma) Transformation (using $\gamma > 1$ to enhance contrast between ridges and valleys).

(ii) Power-law transformation improves the contrast of fingerprint ridges and valleys, making fine ridge details clearer for accurate fingerprint recognition.

(iii) → It would brighten the darker regions.

→ Fine ridge details may become less distinct.

→ Fingerprint matching accuracy may decrease because the ridge contrast is not enhanced as effectively.